

Profile



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About this Portfolio

This portfolio was created to supplement my resume by presenting a selection of projects completed throughout my undergraduate studies in Mechanical Engineering at the University of Ottawa. Throughout my degree, I have worked on projects related to Formula SAE, aerodynamics, powertrain systems, control systems and mechatronics while also gaining industry experience through co-op placements in the powersports and energy sectors. I believe that hands-on engineering projects are one of the most important aspects of undergraduate studies since they provide practical experience beyond the classroom environment. The following pages highlight some of the technical, analytical and multidisciplinary projects I have developed throughout my studies.

Powertrain Lead – Formula uOttawa

Objective

As the Powertrain lead of the Formula uOttawa team, my role was to build and manage a small team of students to support the iterative development of the vehicle powertrain system around the team objectives. From the beginning of the season, the primary goals focused on reliability, fit and finish and testing, and validation. Starting from a team of only three members, achieving these objectives required recruiting, mentoring and coordinating additional students across multiple stages of research, design, manufacturing and system integration.



Leadership & Technical Development

Project Management:

- Hosted regular subteam meetings to coordinate project progress, technical discussions, and task allocation reaching over 10 active subteam members.
- Maintained member retention within a volunteer-based student engineering team through mentorship and active project involvement.
- Introduced new members to ongoing projects through subsystem architecture presentations, technical design overviews and participation in a dyno session.
- Helped students develop practical experience in vehicle assembly and troubleshooting.
- Organized subsystem projects and manufacturing timelines to meet internal deadlines.

Testing and Validation:

- Increased peak power from 79hp to 82.5hp through iterative and tuning designs.
- Increased data acquisition through sensor implementation and organized testing plans.
- Validated designs using engine data, EGT readings and lambda targets.

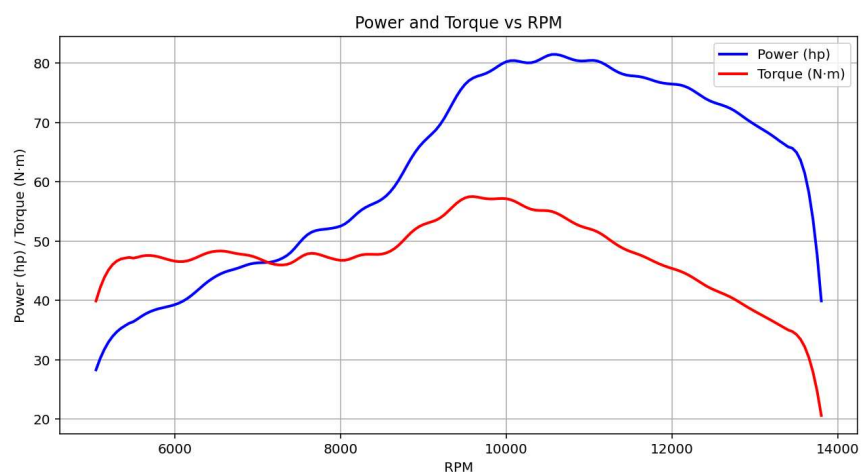


Skills

Software: SolidWorks, Ansys Fluent, Realis WAVE, MoTec M1 Tune, Microsoft Office

Engineering: Powertrain Development, Engine Tuning, Vehicle Testing, Design Reviews

Leadership: Team Coordination, Mentorship, Project Planning, Technical Communications



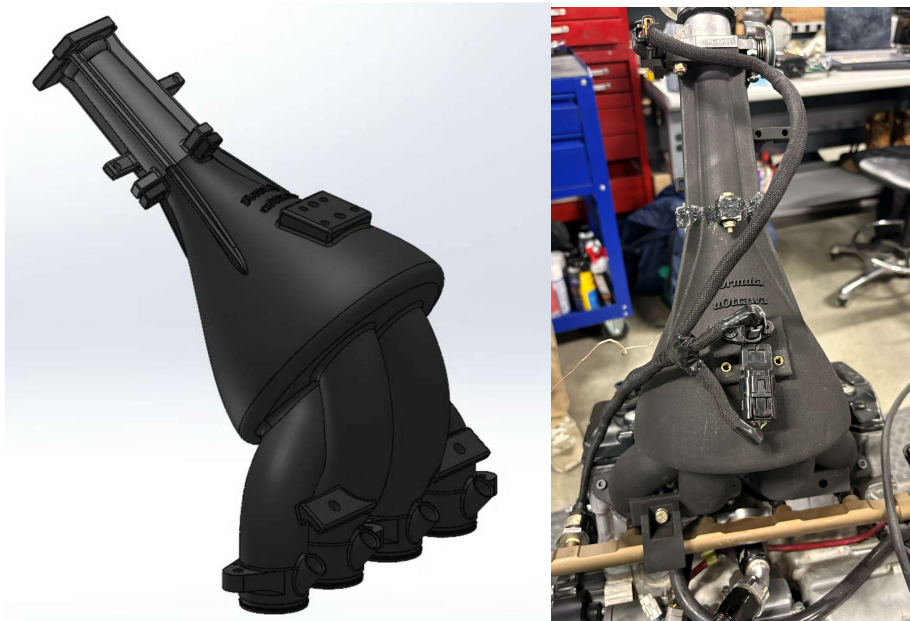
Results and Lessons Learned

- Contributed to the increase in the team's overall Formula SAE competition ranking from 83rd to 41st through improvement in subsystem reliability, testing, and overall fit and finish.
- Contributed to a significant increase in design event scoring through improved engineering justification, subsystem validation, and report writing.
- Increased competitive potential of the team through improved reliability and DAQ implementation supporting strong potential for a top-20 performance in the next season.

Intake Manifold – Formula uOttawa IC

Objective

The objective of this project was to design and optimize an intake manifold for a Suzuki GSX-R600 engine targeting improved volumetric efficiency, airflow distribution, and overall engine performance. With the FSAE rulebook mandating a restrictor and packaging constraints, the intake manifold design required analytical and simulation-based methods for runner sizing, plenum design and flow behaviour optimization within the permissible design envelope.



Intake System Development

Intake Manifold Design:

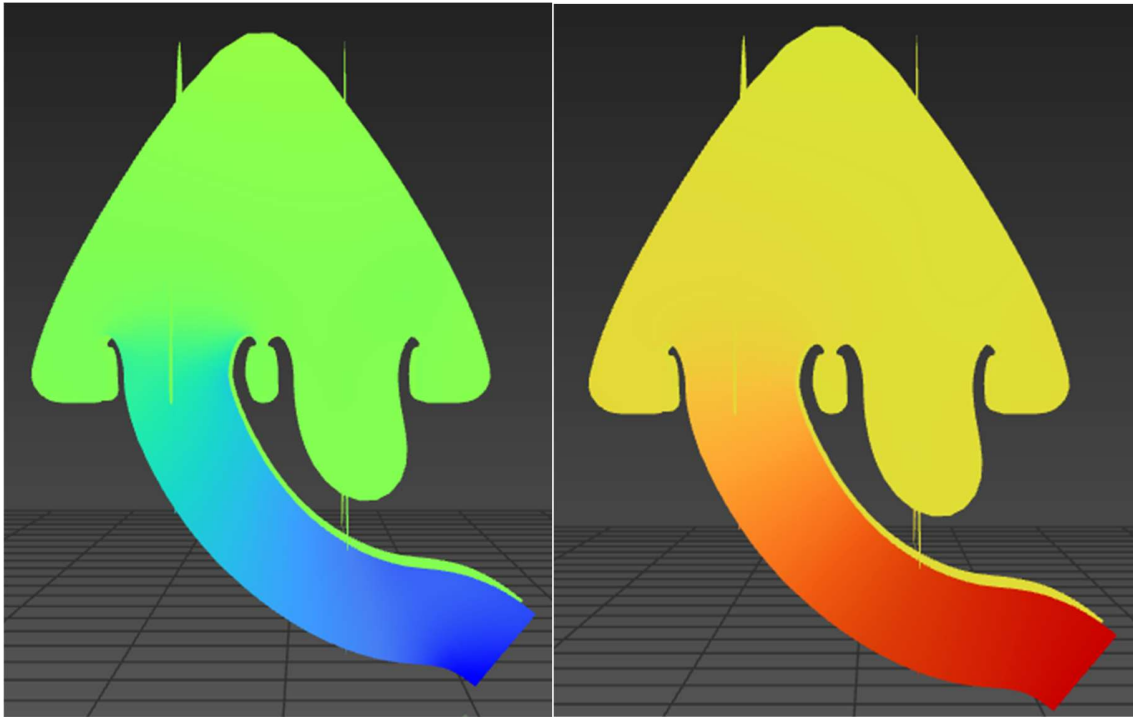
- Square configuration for even airflow distribution and volumetric efficiency targets.
- Runner length tuned for 10 000 RPM using Helmholtz resonation wave theory.
- Plenum sizing targeted improved throttle response. Runner inlet designed as bell mouth.

Engineering Analysis:

- Realis WAVE simulation used for design iteration comparing VE and torque graphs.
- Steady-state CFD used to analyse flow separation and pressure recovery in the diffuser.
- Transient CFD performed assuming maximum VE conditions to validate airflow distribution by measuring pressure averages through the runner outlet.

Validation Methods:

- Exhaust gas temperature data extracted from each exhaust runner measuring 4% discrepancy between the outer and middle cylinders showing improved distribution.
- VE tune curve showing a major spike at 10 000 RPM with the largest effect at full throttle.



Skills

Software: SolidWorks, Ansys Fluent, Realis WAVE

Engineering: Design, Fluid Mechanics, Simulation Analysis, Engine Airflow Analysis

Manufacturing & Validation: Additive manufacturing, Packaging, Testing and Validation

Results and Lessons Learned

- Successfully applied a complete engineering design processes (requirements, design, analysis, manufacturing and validation) to produce a validated intake manifold.
- Demonstrated the influence of each intake system component on the volumetric efficiency, airflow distribution and throttle response.
- Gained experience creating testing plans to validate analytical results for a design.

Aerodynamics Parametrisation – FSAE EV CAPSTONE

Objective

As a part of a five-member capstone team, the objective of this project was to parametrize and optimize a Formula SAE EV under varying conditions to meet performance targets. My role focused on the aerodynamic subsystem, where I developed analytical aerodynamic and optimization models to reach target downforce, balance, cooling, and subsystem integration based on user-defined inputs and vehicle constraints defined in the competition rulebook.



Aerodynamics Development

Aerodynamics Design:

- Multi-element wing designs to minimize flow separation and generate high downforce.
- Undertray tunnels which produce the highest lift to drag ratio due to relatively low drag.
- Sidepods creating low-pressure regions behind the radiators to promote increased airflow.

Engineering Analysis:

- Performed vehicle dynamics modelling to determine downforce targets in cornering.
- Utilized multi-element airfoil theory and induced velocity to estimate downforce.
- Analysed pressure loss, Bernoulli's principle and energy conservation for the sidepods.

Optimization:

- User-defined inputs of car mass, lateral acceleration targets and average cornering radius.
- Iterative optimization approach used to vary angles of attack to achieve downforce targets.

$$w_{ij} = \frac{\Gamma_j}{2\pi} \frac{\Delta x}{\Delta x^2 + \Delta y^2}$$

$$\varepsilon_{ij} = \frac{w_{ij}}{V_\infty c_{l,j}}$$

$$\begin{bmatrix} 1 & a_1\varepsilon_{12} & a_1\varepsilon_{13} \\ a_2\varepsilon_{21} & 1 & a_2\varepsilon_{23} \\ a_3\varepsilon_{31} & a_3\varepsilon_{32} & 1 \end{bmatrix} \begin{bmatrix} c_{l,1} \\ c_{l,2} \\ c_{l,3} \end{bmatrix} = \begin{bmatrix} a_1\eta_1 \\ a_2\eta_2 \\ a_3\eta_3 \end{bmatrix}$$

Equations included: Multi-element Airfoil Theory Induced Velocity, Influence Factor, Element Coefficient of Lift Matrix

Skills

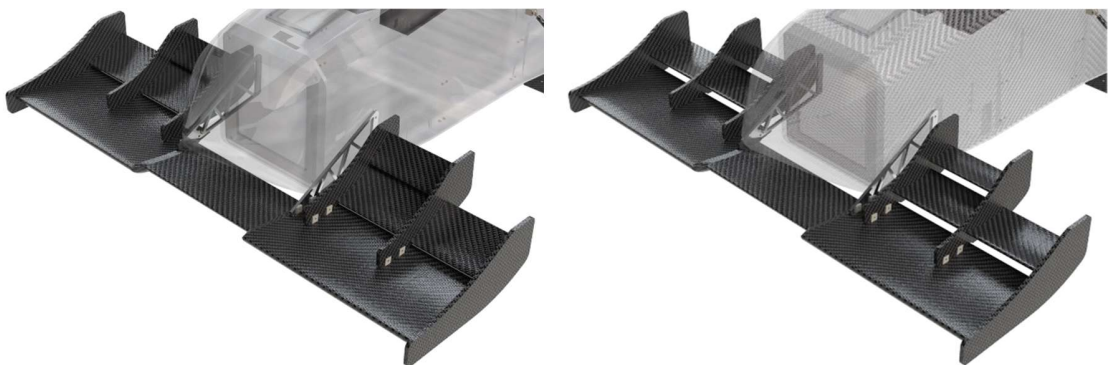
Software: SolidWorks, MATLAB, Microsoft Office (Excel, Word)

Engineering: Design, Modelling, Aerodynamics Analysis, Optimization, Parametrization

Professional skills: Team Collaboration, Report Writing, Technical Communication, Flowcharts

Results and Lessons Learned

- Demonstrated the importance of aerodynamic balance, cooling flow management, and subsystem integration in overall vehicle performance.
- Identified the limitations of simplified aerodynamic modelling methods when applied to complex multi-element flow interactions and ground-effect behaviour.
- Developed high-quality report writing and organizing skills through collaborative reports.



Quadcopter – ELG3736 Course Project

Custom Arduino-Based Mechatronics Drone System

Objective

The objective was to develop a fully functional quadcopter capable of stable flight and real-time environmental monitoring. The project requirements from the course involved the implementation of three sensors while prohibiting the use of pre-built kits or drone-specific flight controllers. Instead, it was required to use a microcontroller-based architecture with wireless communication from the ground up.

Key Contributions

Mechanical:

- Designed the chassis, mounting system and propellers using SolidWorks.
- Designed lightweight components optimized for 3D-printed and system packaging.
- Performed iterative design for chassis, mounting and propeller sizing and tolerancing.

Electrical:

- Developed a custom dual Arduino-based control and wireless communication system.
- Integrated ultrasonic, pressure/altitude, and gyroscopic sensors into the drone electronics.
- Designed and assembled the onboard circuitry, including power distribution.

Testing and Validation:

- Performed iterative subsystem prototyping by testing, debugging, and troubleshooting.
- Validated structural integrity and electrical functionality with incremental testing.
- Identified microcontroller processing limitations through full-system integration.

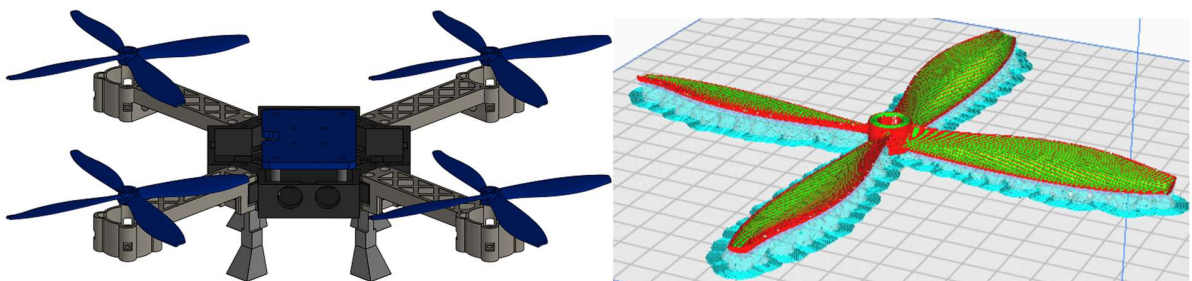


Figure 1: CAD Model and Propeller 3D-Print Slice

Skills

Software: SolidWorks, Arduino IDE, Simulink (MATLAB)

Manufacturing & Validation: Additive manufacturing, Rapid prototyping, Testing and validation

Electrical: Circuit design, Wireless communication, Sensor integration

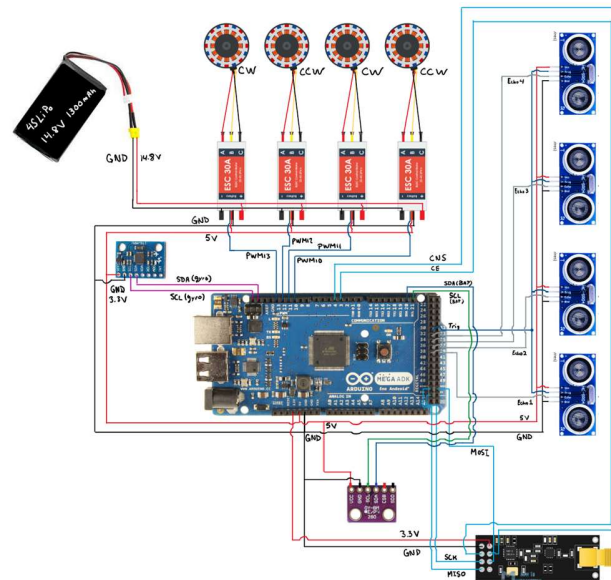


Figure 2: Onboard Electronics Circuit Diagram

Results and Lessons Learned

Wireless communication and sensor integration were successfully implemented through the project development process. Full-system testing revealed processing limitations of the Arduino platform when integrating the gyroscopic sensor which highlights the importance of processing efficiency and subsystem optimization in the mechatronics systems.



Figure 3: Completed Physical Model Controller and Quadcopter

Inverted Pendulum – Course Project

Objective

The objective of this project was to design an inverted pendulum system to develop a deeper understanding of PID control in the context of a control systems course. The goal was to balance the pendulum through a closed-loop feedback system by tuning the proportional, integral, and derivative gains until the system could maintain a stable upright position.

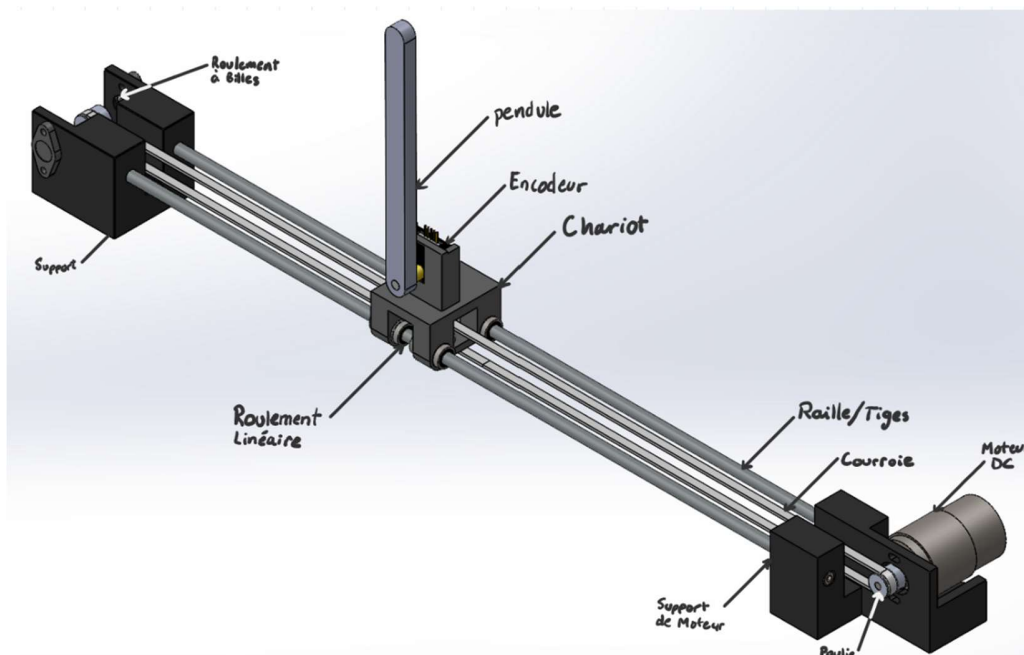


Figure 4: Inverted Pendulum CAD Model

Key Contributions

Mechanical:

- Designed a rail-guided cart system using SolidWorks.
- Mechanically constraining the system to 2D to simplify the pendulum motion and controls.
- Implemented belt tensioning and bearing retention features.

Controls & Electrical:

- Integrated an Arduino control system with encoder feedback and a NEMA DC motor.

Testing and Validation:

- Analysed the PID controls behaviours using Simulink closed-loop system modelling.
- Participated in the tuning of the PID gains through iterative physical testing.

Skills

Software: SolidWorks, Arduino IDE, Simulink (MATLAB)

Manufacturing & Validation: Additive manufacturing, Rapid prototyping, Testing and validation

Controls & Electrical: PID Control, Circuit design, PWM Motor Control

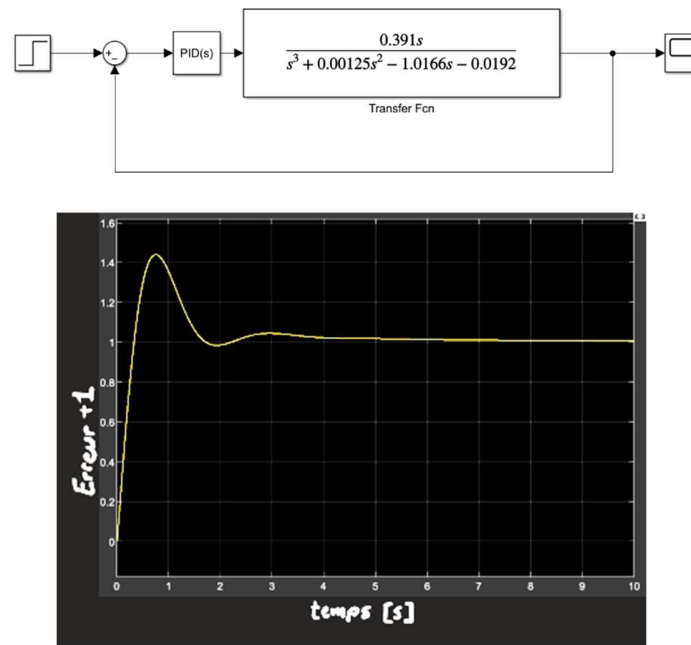


Figure 5: Simulink Closed Loop Simulation with Step Response Plot

Results and Lessons Learned

- Successfully developed a closed-loop PID control system capable of balancing the inverted pendulum for extended periods under stable operating conditions.
- Strengthened practical experience in control systems, Simulink modelling, and mechatronics design.



Figure 6: Inverted Pendulum Assembly in Starting Position